Empirical Applications (2): Alien Species Invasions

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Beyond Linearity: Modeling a Bimodal Network of Alien Species Invasions

Preliminarities

The following section aims to load the necessary packages and import the data.

```
load(file="input/input-alien.RData")
```

First records

We firstly show how data arrives from the original source (Seebens, 2017).

```
head(FR[,c("LifeForm", "Taxon",
           "Region", "FirstRecord",
           "Source")])
```

```
##   LifeForm                     Taxon           Region FirstRecord
## 1    Algae    Acanthophora muscoides           Turkey        1986
## 2    Algae Acanthophora nayadiformis           Cyprus        1997
## 3    Algae Acanthophora nayadiformis           Turkey        1970
## 4    Algae    Acanthophora spicifera Hawaiian Islands        1952
## 5    Algae    Acetabularia caliculus           Israel        1943
## 6    Algae    Acetabularia caliculus            Spain        1957
##                      Source
## 1      Cinar et al. (2005)
## 2                   DAISIE
## 3      Cinar et al. (2005)
## 4 Carlton & Eldrege (2009)
## 5                   DAISIE
## 6                   DAISIE
```

These data involve 17 life forms. For each record, we are able to identify the species (`Taxon`) of a certain lifeform (`Lifeform`) that is invading a region (`Region`) in which it was not native and was not seen before the reported year (`FirstRecord`). The Alien Species First Records Database collects several sources: the source of each FR is reported in the column `Source`.

We recall that a first record is the year (t) when a (non-native) species s first invades region r. The following dataset, FR, already contains information we need to interpret first records as relational events (s, r, t). Particularly, the sender set is composed of the unique recorded species, while regions of the world (countries and islands) form the receiver set. The time window spans from $t_0 = 1880$ to $T = 2005$.

```
head(first_records[,c("year","lf",
                      "species",
```

```
                        "region")])
```

```
##       year      lf        species         region
## 19610 1880 Mammals  Axis porcinus        Denmark
## 19702 1880 Mammals  Canis latrans United Kingdom
## 19837 1880 Mammals Cervus elaphus         Russia
## 19907 1880 Mammals      Dama dama        Algeria
## 19922 1880 Mammals      Dama dama           Fiji
## 19933 1880 Mammals      Dama dama          Libya
```

Native range

A species' native range (NR) is the collection of areas where it is indigenous. Slightly more liberally, we refer to native range as the set of sites where a species was already present before the start of the analysis period, which in this context is 1880. `native` object contains the FRs, involving insects, that occurred before 1880. An example below,

Why do we need the native range? Because the relational event is non-recurrent, namely, the event first record, for its definition, occurs only once. This also means that the risk set is reducing over time: once the event is observed, it cannot be sampled anymore as a non-event. But also because covariates are endogenous and at each time, the first time included, rely on the knowledge of which first records already occurred and this must be possible at the beginning of the analysis as well. D1.5: Inspect the object first_records. Describe this relational event network and identify why it can be considered a subset of the previous one.

```
native[sample(1:nrow(native), 10),c("species", "region")]
```

```
##                         species        region
## 206292    Oryctolagus cuniculus         Tonga
## 37985                  Dama dama        Jordan
## 38735          Felis silvestris       Burundi
## 65745            Talpa europaea       Belgium
## 34940          Acinonyx jubatus      Tanzania
## 41541        Loxodonta africana Guinea-Bissau
## 42339              Mungos mungo     Swaziland
## 65868            Tayassu pecari     Nicaragua
## 65903  Tragelaphus strepsiceros    Mozambique
## 58858              Panthera leo          Iran
```

In Juozaitienė et al (2023), only a subset of these first records has been analyzed, including first records of insects, plants, birds and mammals between 1880 and 2005, involving species for which the native range was known. The set of analyzed FRs is contained in the object subset. In this tutorial, in order to make the analysis easy to reproduce, we focus on mammals only. `first_records` contains the mammal FRs of interest.

```
nrow(first_records)
```

```
## [1] 799
```

```
range(first_records$year)
```

```
## [1] 1880 2005
```

```
# Define sender set - species
spec <- unique(first_records$species)
(s <- length(spec))
```

```
## [1] 186
```

```r
# Define receiver set - regions
reg.lf <- unique(first_records$region)
(r <- length(reg.lf))
```

```
## [1] 153
```

```r
reg <- colnames(data_distance)
```

Invaded regions

This type of function allows us to identify particular instances in the previous event history (specifically, past events involving the same species) and to extract attributes of interest (e.g., which regions were invaded). This information can then be aggregated into a covariate. Since this covariate is a function of the past network of events, the resulting covariate is endogenous. Since we do not store the time of these events - but only the receivers - the covariate will not be a function of the internal time.

```r
invaded.regions <- function(sp.n, r.n, y, native, first_records){

  # Convert input arguments to numeric type if not already
  sp.n <- as.numeric(sp.n)
  r.n <- as.numeric(r.n)
  y <- as.numeric(y)

  # Get unique combinations of species number and region number from native data
  t <- unique(as.vector(subset(native, sp.num == sp.n, r.num)))

  # Get region numbers from first records data where species number matches
  pr <- as.vector(subset(first_records, sp.num == sp.n, r.num))

  # If the invasion is both present in first records data and in native range
  # consider the former as actual piece of information
  t <- setdiff(t, pr)

  # Find indices of first records occurring before end date for the species
  set.sp <- which(first_records$sp.num == sp.n & first_records$year < y)

  # Combine regions in native range with regions in first records before date
  t <- na.omit(c(t, first_records$r.num[set.sp]))

  # Do not consider the involved region
  inv <- unlist(setdiff(t, r.n))

  # Return invaded regions
  return(inv)
}
```

Creating the case-control dataset

The dynamic network of FRs consists of two sets of nodes: the set of species and the set of regions. Additionally, in the dataset, we refer to a numerical formulation of species and regions, which correspond to the set of unique observed species in the data and the set of regions reported in the matrix of geographical distances, respectively.

To perform inference using the nested case-control partial likelihood with m=1, we construct the case-control dataset, which includes, for each first record, the dyad involved in the observed event and that involved in

the non-event randomly selected from the risk-set. It is essential to emphasize that in this scenario, due to the non-recurrent nature of the relational event, the risk set diminishes over time. Once the event occurs, it cannot occur again and thus, it is excluded from the dyads at risk.

In order to address the ties in the event sequence, we exclude, for each event, all the first records that occurred in the same year from being considered as potential non-events.

```
creating_case_control_dataset <- function(first_records,
                                          spec, reg.lf, reg,
                                          seed=1234
){

  reg.lf.num <- match(reg.lf, reg)
  s <- length(spec)
  r <- length(reg.lf)

  ## POSSIBLE (S,R) INTERACTIONS ####
  # Initialize a vector to keep track of the risk set size over time
  at.risk <- NULL
  # Create a matrix to indicate possible (species, region) interactions
  alien.occ <- matrix(0, nrow = s, ncol = r)
  rownames(alien.occ) <- spec
  colnames(alien.occ) <- reg.lf
  for (n.sp in 1:s){
    # Identify native regions for each species
    nat.id <- unique(native$r.num[native$sp.num == n.sp])
    # Identify regions where the species is not native
    possible.to <- setdiff(reg.lf.num,nat.id)
    # Mark these regions as possible invasion sites for the species
    alien.occ[n.sp,reg[possible.to]] <- 1
  }

  ## COLLECTING INFORMATION ####
  dat.gam <- data.frame(matrix(NA, nrow=nrow(first_records),ncol=6))
  colnames(dat.gam) <- c("y", "year",
                         "sp1", "r1",
                         "sp2", "r2")
  # The response is fixed and equal to 1
  dat.gam[,1] <- rep(1, nrow(first_records))

  set.seed(seed)
  # For each FR:
  for (i in 1:nrow(first_records)){

    ### INFORMATION CONCERNING THE EVENT ####
    # year of the invasion event
    dat.gam[i,2] <- year <- first_records[i,"year"]
    # invading species
    dat.gam[i,3] <- s.ev <- first_records[i,"species"]
    # invaded country
    dat.gam[i,4] <- r.ev <- first_records[i,"region"]

    ### POSSIBLE EVENTS ####
    # Events occurred at the same time of the considered event
    # are removed from the risk set
```

```r
    sub_stp <- first_records[first_records$year==year,
                             c("species", "region")]
    ni <- nrow(sub_stp)
    for (j in 1:ni){
      # Mark these (species, region) pairs as not at risk
      alien.occ[sub_stp[j,1],sub_stp[j,2]] <- 0
    }
    at.risk <- c(at.risk, sum(alien.occ==1))

    ### SAMPLING THE NON-EVENT ####
    sr.nv<-sample(which(alien.occ!=0),1)
    # species non-event
    dat.gam[i,5] <- s.nv <- spec[(sr.nv-1)%%s+1]
    # region non-event
    dat.gam[i,6] <- r.nv <- reg.lf[(sr.nv-1)%/%s+1]
  }

  return(dat.gam)
}
```

```r
set.seed(1234)
dat.gam <- creating_case_control_dataset(first_records,
                                         spec, reg.lf, reg)
dat.gam$sp1.num <- match(dat.gam$sp1, spec)
dat.gam$r1.num <- match(dat.gam$r1, reg)
dat.gam$sp2.num <- match(dat.gam$sp2, spec)
dat.gam$r2.num <- match(dat.gam$r2, reg)

head(dat.gam[c("year","sp1", "r1","sp2", "r2")])
```

```
##   year            sp1             r1                 sp2                r2
## 1 1880  Axis porcinus        Denmark     Rusa timorensis            Poland
## 2 1880  Canis latrans United Kingdom  Spermophilus major Clipperton Island
## 3 1880 Cervus elaphus         Russia       Vulpes corsac          Bulgaria
## 4 1880      Dama dama        Algeria   Macaca nemestrina Clipperton Island
## 5 1880      Dama dama           Fiji    Atelerix algirus           Namibia
## 6 1880      Dama dama          Libya Rupicapra rupicapra             Palau
```

```r
head(dat.gam[c("year","sp1.num", "r1.num","sp2.num", "r2.num")])
```

```
##   year sp1.num r1.num sp2.num r2.num
## 1 1880       1     32      65     44
## 2 1880       2     14     111    271
## 3 1880       3     13      97      6
## 4 1880       4    182     181    271
## 5 1880       4    107      30    173
## 6 1880       4    217      64    113
```

Covariate computation

Climatic dissimilarity

`data_temperature` contains the names of the countries in the first column and the average values of near-surface air temperature in the second column (Watanabe et al., 2011). The covariate depends on both the sender and the receiver of the event and can therefore be considered a dyad-level covariate. It is time-

dependent but does not incorporate information about internal time. It is endogenous, as it relies on the regions previously invaded by the species, but it also incorporates exogenous information.

```
climatic_dissimilarity <- function(sp.n, r.n, y, native, first_records,
                                   reg, data_temperature){

  # Convert input arguments to numeric type if not already
  sp.n <- as.numeric(sp.n)
  r.n <- as.numeric(r.n)
  y <- as.numeric(y)

  # Find regions invaded by the species before the current time
  inv <- invaded.regions(sp.n = sp.n,
                         r.n = r.n,
                         y = y,
                         native = native,
                         first_records = first_records)

  # Consider the minimum absolute difference in temperature
  # between the region of interest and those already invaded
  avg_temp_invaded <- data_temperature[data_temperature[,1] %in% reg[inv], 2]
  # Get the average temperature value for the region of interest
  avg_temp_interest <- data_temperature[r.n, 2]
  # Find the minimum absolute difference in temperature
  dt.value <- min(abs(avg_temp_invaded - avg_temp_interest))

  # Return the calculated climatic dissimilarity value
  return(dt.value)
}

dat.gam$dt1 <- apply(dat.gam[,c("sp1.num", "r1.num", "year")], 1,
                     function(x) climatic_dissimilarity(sp.n = x[1],
                                                        r.n = x[2],
                                                        y = x[3],
                                                        native = native,
                                                        first_records =
                                                          first_records,
                                                        reg = reg,
                                                        data_temperature =
                                                          data_temperature))
dat.gam$dt2 <- apply(dat.gam[,c("sp2.num", "r2.num", "year")], 1,
                     function(x) climatic_dissimilarity(sp.n = x[1],
                                                        r.n = x[2],
                                                        y = x[3],
                                                        native = native,
                                                        first_records =
                                                          first_records,
                                                        reg = reg,
                                                        data_temperature =
                                                          data_temperature))
dat.gam$dt = dat.gam$dt1 - dat.gam$dt2
head(dat.gam)
```

```
##   y year            sp1             r1                  sp2                r2
## 1 1 1880  Axis porcinus        Denmark     Rusa timorensis            Poland
```

```
## 2 1 1880  Canis latrans United Kingdom  Spermophilus major Clipperton Island
## 3 1 1880 Cervus elaphus         Russia       Vulpes corsac          Bulgaria
## 4 1 1880      Dama dama        Algeria   Macaca nemestrina Clipperton Island
## 5 1 1880      Dama dama           Fiji    Atelerix algirus           Namibia
## 6 1 1880      Dama dama          Libya Rupicapra rupicapra             Palau
##   sp1.num r1.num sp2.num r2.num        dt1        dt2          dt
## 1       1     32      65     44  2.0065592 13.6226336 -11.6160744
## 2       2     14     111    271 11.0616861 17.7612506  -6.6995645
## 3       3     13      97      6  5.9904097  1.5396019   4.4508078
## 4       4    182     181    271  1.1605766  0.6140593   0.5465172
## 5       4    107      30    173  0.5544317  1.2652227  -0.7107910
## 6       4    217      64    113  0.3991832 10.4835401 -10.0843569
```

Trade

In the existing literature, international trade has been recognized as a key factor in explaining the spread of alien species. Here, we consider trade as the yearly commerce between already-invaded territories and the involved region. The source information, derived from Barbieri et al. (2009), which reports trade flows among countries, is contained in `data_trade`.

A few notes on data pre-processing: - Negative values of trade are considered equal to 0. - When a pair of nations is never mentioned in the source data, the trade volume is presumed to be 0, based on the idea that if a value is never recorded, it should be negligible. - When the pair is mentioned but there is a non-imputed gap at the time of interest, the trade flow is evaluated based on the latest available year for which information is available. - Because trade varies by orders of magnitude, we take a log-transformation for the analysis.

```r
t <- which(data_trade$transfer < 0)
data_trade$transfer[t] <- 0

trade.funct <- function(inv, r.n, y, reg, data_trade){

  # Convert input arguments to numeric type if not already
  r.n <- as.numeric(r.n)
  y <- as.numeric(y)

  # Check if there are already invaded countries
  if(length(inv)!=0){

    # Find rows that involve invaded countries as sending trade
    u <- which(data_trade$FromRegion %in% reg[inv])
    # Find rows that involve region of interest as receiving trade
    v <- which(data_trade$ToRegion == reg[r.n])
    # Find the intersection of the two sets
    w <- intersect(u,v)
    x <- data_trade[w,]
    # Consider the trade instances occurred before or at the time of interest
    x <- x[x$year<=y,]
    trade_value <- NULL
    # If there are rows in the filtered dataset
    if(nrow(x)>0){
      # For each invaded country, the maximum year is recorded
      o <- aggregate(x$year, list(x$FromRegion), FUN=max)
      # For each of them, the corresponding transfer is stored
      for (o.i in 1:nrow(o)){
        trade_value <- c(trade_value,
```

```
                        x$transfer[x$FromRegion==o[o.i,1] &
                                   x$year==o[o.i,2]])}
    }
  } else {
    # If there are not already invaded countries, trade is set equal to 0
    trade_value <- 0
  }
  # Compute the log-transformed sum of trade values (with an added constant 1)
  log_trade.value <- ifelse(length(trade_value)>0,
                            log(sum(trade_value, na.rm =T)+1),0)

  # Return the computed log-transformed trade value
  return(log_trade.value)
}

log_trade <- function(sp.n, r.n, y, native, first_records, reg, data_trade){

  inv <- invaded.regions(sp.n = sp.n,
                         r.n = r.n,
                         y = y,
                         native = native,
                         first_records = first_records)

  log_trade.value <- ifelse(r.n==match("USACanada", reg),
                            mean(c(trade.funct(inv = inv,
                                               r.n = match("United States",reg),
                                               y = y,
                                               reg = reg,
                                               data_trade = data_trade),
                                   trade.funct(inv = inv,
                                               r.n = match("Canada",reg),
                                               y = y,
                                               reg = reg,
                                               data_trade = data_trade))),
                            trade.funct(inv = inv,
                                        r.n = r.n,
                                        y = y,
                                        reg = reg,
                                        data_trade = data_trade))

  return(log_trade.value)
}

dat.gam$tr1 <- apply(dat.gam[,c("sp1.num", "r1.num", "year")], 1,
                     function(x) log_trade(sp.n = x[1],
                                           r.n = x[2],
                                           y = x[3],
                                           native = native,
                                           first_records =
                                             first_records,
                                           reg = reg,
                                           data_trade = data_trade))
dat.gam$tr2 <- apply(dat.gam[,c("sp2.num", "r2.num", "year")], 1,
```

```
                     function(x) log_trade(x[1], x[2], x[3],
                                           native = native,
                                           first_records =
                                             first_records,
                                           reg = reg,
                                           data_trade = data_trade))
dat.gam$tr = dat.gam$tr1 - dat.gam$tr2
```

Distance

We compute distance for the dyad (species-region) as the shortest distance among the regions in which the species is already present at that time. Distance is computed referring to the closest borders and consequently is equal to zero for neighbours. Source data for this driver consists of the R package geosphere (Hijmans et al., 2017).

`data_distance` is a matrix with rownames and colnames referring to the regions. The corresponding element in the matrix reports the distance between the two.

```
log_distance <- function(sp.n, r.n, y, native, first_records, data_distance){

  # Convert input arguments to numeric type if not already
  sp.n <- as.numeric(sp.n)
  r.n <- as.numeric(r.n)
  y <- as.numeric(y)

  # Find regions invaded by the species before the current time
  inv <- invaded.regions(sp.n = sp.n,
                         r.n = r.n,
                         y = y,
                         native = native,
                         first_records = first_records)

  # Consider the logarithm of the minimum distance
  # between the region of interest and those already invaded
  log_dist.value <- log(min(data_distance[r.n, inv])+1)

  # Return the calculated distance
  return(log_dist.value)
}

dat.gam$d1 <- apply(dat.gam[,c("sp1.num", "r1.num", "year")], 1,
                    function(x) log_distance(x[1], x[2], x[3],
                                             native = native,
                                             first_records =
                                               first_records,
                                             data_distance = data_distance))
dat.gam$d2 <- apply(dat.gam[,c("sp2.num", "r2.num", "year")], 1,
                    function(x) log_distance(x[1], x[2], x[3],
                                             native = native,
                                             first_records =
                                               first_records,
                                             data_distance = data_distance))
dat.gam$d = dat.gam$d1 - dat.gam$d2
```

Model fitting

REM w/ linear effect

We estimate the linear fixed effect of climatic dissimilarity and interpret the resulting estimate.

```
gam_dt.only <- gam(y ~ dt - 1,
    family="binomial"(link = 'logit'), data=dat.gam)
```

```
summary(gam_dt.only)
```

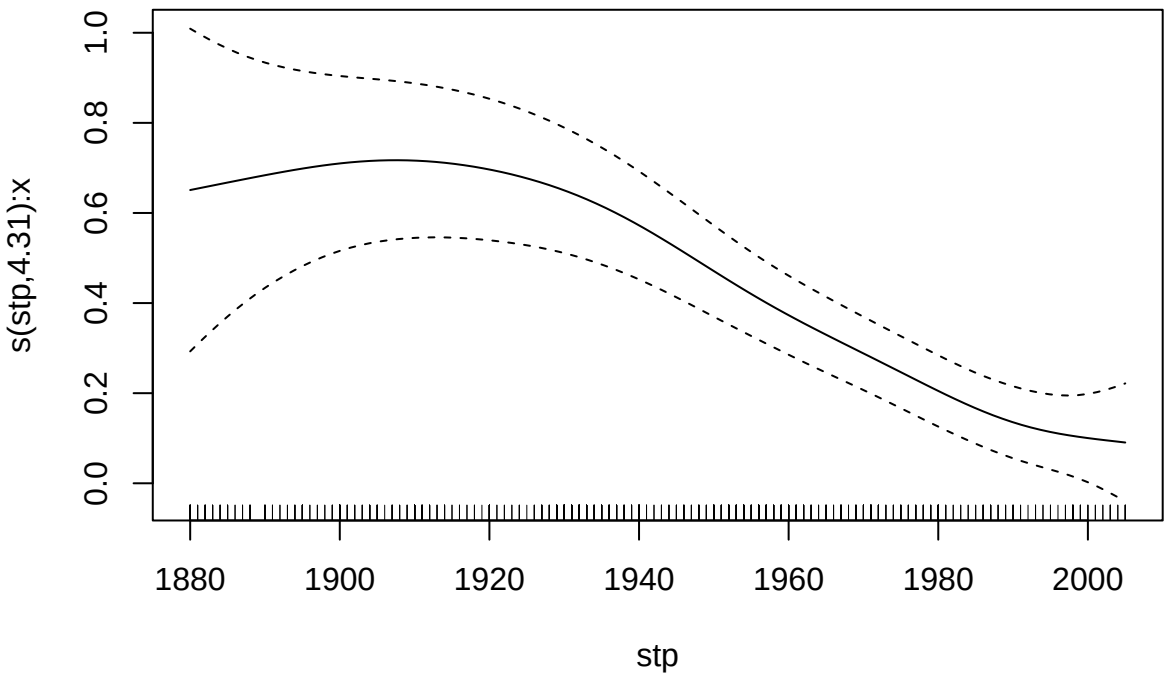
```
##
## Family: binomial
## Link function: logit
##
## Formula:
## y ~ dt - 1
##
## Parametric coefficients:
##    Estimate Std. Error z value Pr(>|z|)
## dt -0.15249    0.01621  -9.405   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## R-sq.(adj) =   -Inf   Deviance explained = -Inf%
## UBRE = 0.22131  Scale est. = 1         n = 799
```

The coefficient for Climatic Dissimilarity, which remains constant over time, is negative. This indicates that the rate of invasions tends to increase if the species has already invaded countries with similar temperatures.

REM w/ time-varying effect

Let `x.ev` and `x.nv` be $n \times 1$ vectors of trade covariate evaluated for events & non-events. Then: `x = x.ev - x.nv`. Let `stp` be $n \times 1$ vector of event-times. We fit the model incorporating a time-varying effect for `x`. Interpretation is reported in the manuscript.

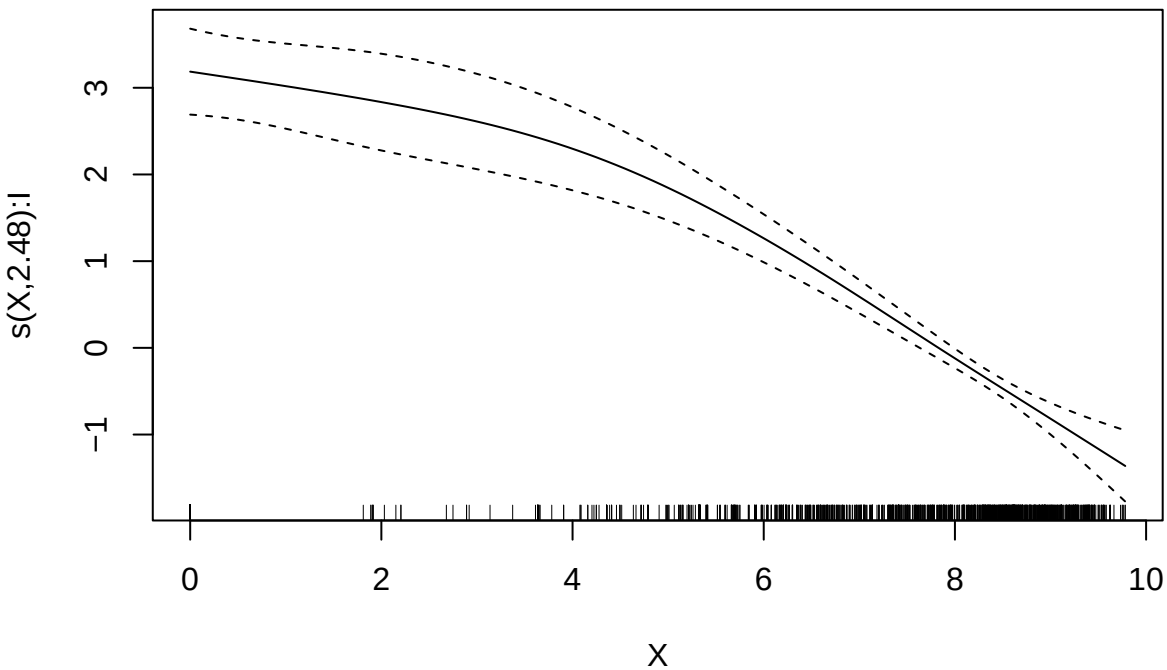
```
# nx1 vector of trade covariate for events
x.ev <- dat.gam$tr1
# nx1 vector of trade covariate for non-events
x.nv <- dat.gam$tr2
x <- x.ev - x.nv # difference between ev and nv

stp <- dat.gam$year ## time

gam_tr.only <- gam(y ~ s(stp, by=x) - 1,
    family="binomial"(link = 'logit'), data=dat.gam)

plot(gam_tr.only)
```



REM w/ non-linear effect

Let `x.ev` and `x.nv` be $n \times 1$ vectors of distance covariate evaluated for events & non-events. Then: `x = x.ev - x.nv`. Let `unit` be $n \times 1$ unit vector. We fit the model incorporating a non-linear effect for `x`. Interpretation is reported in the manuscript.

```
x.ev <- dat.gam$d1
x.nv <- dat.gam$d2
unit <- rep(1, nrow(dat.gam))

X = cbind(x.ev,x.nv)
I = cbind(unit,-unit)

gam_d.only <- gam(y ~ s(X, by=I) - 1,
    family="binomial"(link = 'logit'), data=dat.gam)

plot(gam_d.only)
```



REM w/ random effects

We fit the model incorporating a random effect for species invasiveness.

REMARK: default spline is type for `gam` in `mgcv` consists of thin plate regression spline. Instead, we need to specify that we aim to fit a random effect.

```
sp1 <- dat.gam$sp1
sp2 <- dat.gam$sp2
sp <- factor(c(sp1,sp2))
dim(sp) <- c(length(sp1),2)
```

```
unit <- rep(1, nrow(dat.gam))
I = cbind(unit,-unit)
gam_sp.only <- gam(y ~ s(sp, by=I, bs="re") - 1,
    family="binomial"(link = 'logit'), data=dat.gam)

re.species <- coefficients(gam_sp.only)
names(re.species) <- levels(sp)

# 5 most invasive species
sort(re.species, decreasing = TRUE)[1:5]
```

```
##      Myocastor coypus         Procyon lotor Oryctolagus cuniculus
##              2.351055              2.207190              2.044386
##             Dama dama            Sus scrofa
##              1.846069              1.794064
```

REM w/ all previous effects and model selection

```
unit <- rep(1, nrow(dat.gam))
stp = dat.gam$year
X = cbind(dat.gam$d1,dat.gam$d2)
I = cbind(unit,-unit)
```

```r
gam_complete <- gam(y ~ dt +
                      s(stp, by=tr) +
                      s(X, by=I) +
                      s(sp, by=I, bs="re") - 1,
    family="binomial"(link = 'logit'), data=dat.gam)
```

```r
re_species <- coefficients(gam_complete)[21:length(coefficients(gam_complete))]
names(re_species) <- levels(sp)
sort(re_species, decreasing = TRUE)[1:5]
```

```
##       Procyon lotor       Rattus rattus    Myocastor coypus     Bubalus bubalis
##           0.6399111           0.6171872           0.6041268           0.5697507
## Herpestes javanicus
##           0.4787689
```

```r
AIC(gam_complete)
```

```
## [1] 475.87
```

```r
AIC(gam_d.only)
```

```
## [1] 591.359
```

```r
AIC(gam_tr.only)
```

```
## [1] 762.7196
```

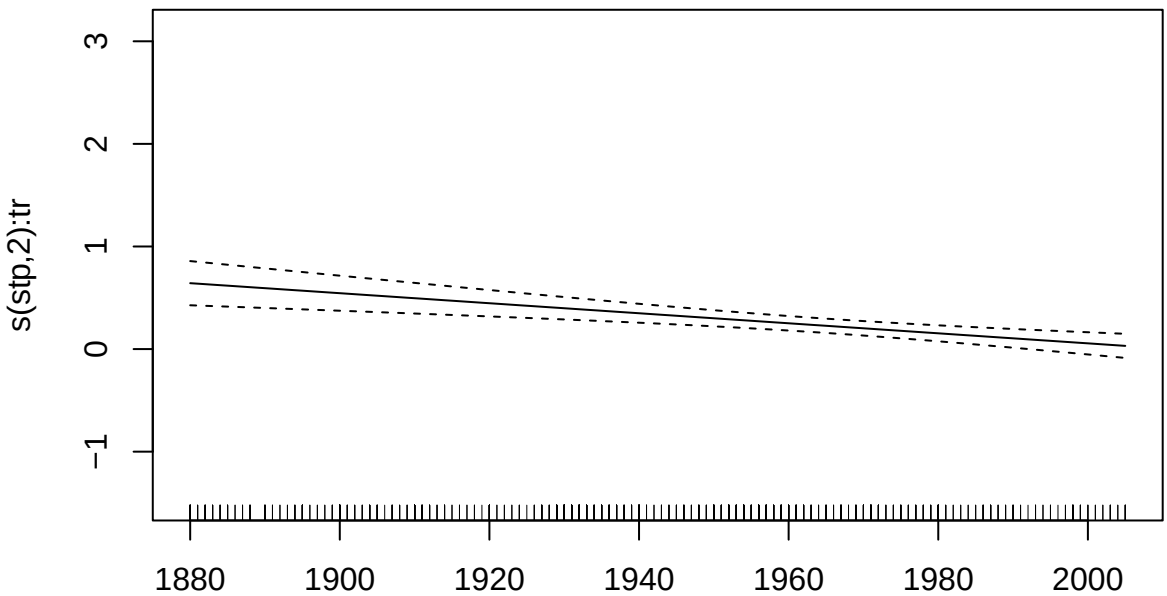
```r
AIC(gam_sp.only)
```

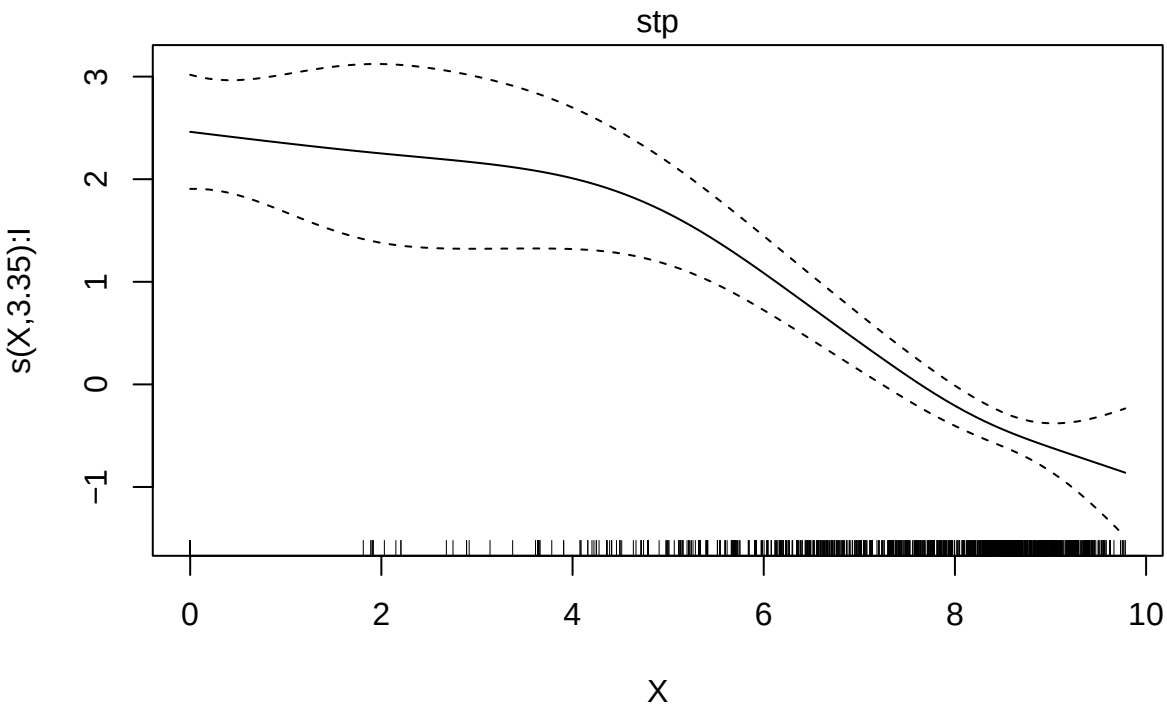
```
## [1] 835.1135
```

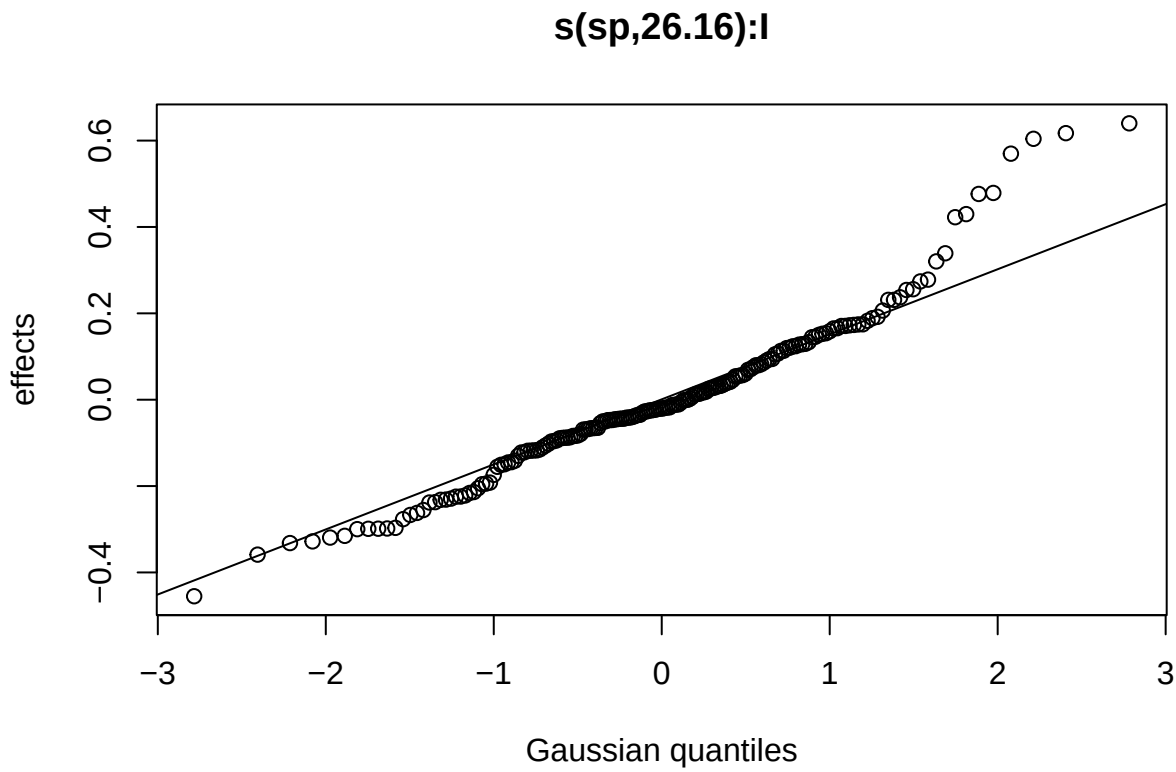
```r
AIC(gam_dt.only)
```

```
## [1] 975.8281
```

```r
plot(gam_complete)
```


s(stp,2):tr
3
2
1
0
−1
1880
1900
1920
1940
1960
1980
2000
stp


s(X,3.35):I
3
2
1
0
−1
0
2
4
6
8
10
X

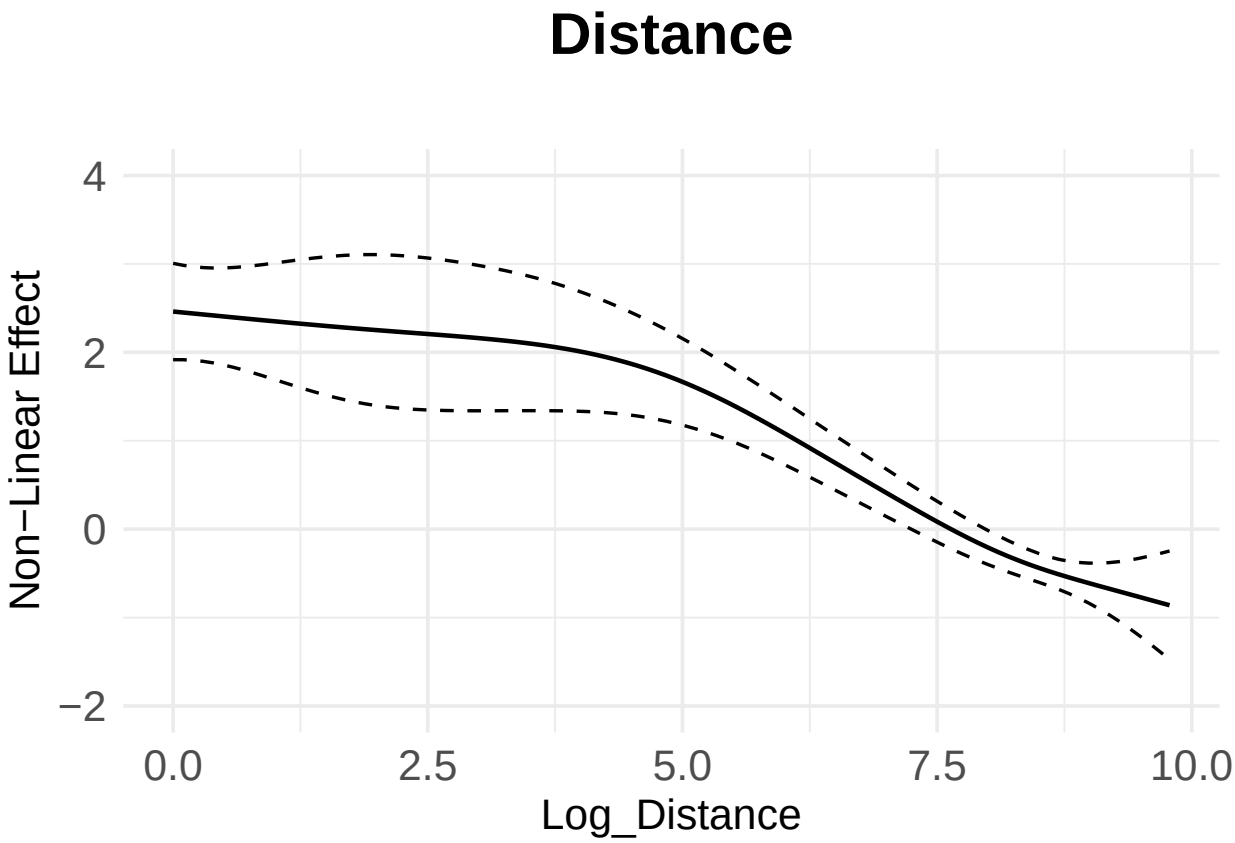



```
create_plots <- function(gam_fit, elements, ranges,
                         x_labs, y_labs, titles, subtitle){

  viz_obj <- getViz(gam_fit)

  plots <- list()

  for (element in elements){

    x_lab <- x_labs[element]
    y_lab <- y_labs[element]
    main_title <- titles[element]

    sm <- plot(sm(viz_obj, select=element))
    data <- sm$ggObj$data

    data$upper <- data$y + 1.96 * data$se
    data$lower <- data$y - 1.96 * data$se

    plots[[element]] <- ggplot(data, aes(x = x, y = y)) +
      geom_line(color = "black", linewidth = 0.8) +
      geom_line(aes(y = upper), linetype = "dashed", color = "black",
                linewidth = 0.6) +
      geom_line(aes(y = lower), linetype = "dashed", color = "black",
                linewidth = 0.6) +
      labs(
        x = x_lab,
        y = y_lab,
        title = main_title,
```

```r
        subtitle = subtitle,
      ) +
      theme_minimal(base_size = 14) +
      theme(
        plot.title = element_text(size = 22, face = "bold", hjust = 0.5),
        plot.subtitle = element_text(size = 18, face = "bold", hjust = 0.5),
        axis.title.x = element_text(size = 16),
        axis.title.y = element_text(size = 16),
        axis.text.x = element_text(size = 16),
        axis.text.y = element_text(size = 16),
        axis.text = element_text(size = 13)
      ) +
      ylim(c(ranges[element,]))

  }

  return(plots)

}

plots <- create_plots(gam_fit = gam_complete,
                              elements = c(1,2),
                              ranges = rbind(c(-0.1,1),
                                             c(-2,4)),
                              x_labs = c("Time",
                                       "Log_Distance"),
                              y_labs = c("Time-Varying Effect",
                                         "Non-Linear Effect"),
                              titles = c("Trade",
                                         "Distance"),
                              subtitle = "")

plots[[1]]
```

Trade
1.00
0.75
0.50
0.25
0.00
Time−Varying Effect
1880
1920
1960
2000
Time

```
plots[[2]]
```


Distance
4
2
0
−2
Non−Linear Effect
0.0
2.5
5.0
7.5
10.0
Log_Distance

```r
pdf("pictures/empirical/alien/TRADE.pdf", width=6, height=6)
plots[[1]]
dev.off()
```

```
## pdf
##   2
```

```r
pdf("pictures/empirical/alien/DISTANCE.pdf", width=6, height=6)
plots[[2]]
dev.off()
```

```
## pdf
##   2
```